Capacity in parallel

Class handout

[*As class starts, Radu passes out a handout with the following case and asks the students to read it out loud as a group, each taking turns to read. Radu gives students some time to ponder and discuss among themselves in smaller groups.*]

Case: Paula and George Expand the Kitchen

Three years after George joined the business, Paula and George's pizza shop had outgrown its original storefront. Demand had grown steadily, the lease was up for renewal, and Paula made a decision: it was time to move. The new location also came with something new—a proper kitchen, with a real pizza oven, deep fryer, and enough floor space for two full cooking stations. The move changed more than just the square footage. The shop no longer sold uncooked, take-home pizzas; with an oven on site, they could bake pizzas and sell them ready to eat. A cooking step was now part of every order.

It also changed how Paula and George worked together. In the old shop, they had passed dough back and forth between two stations, but now each could run a complete pizza line on their own—tossing the dough, topping it, and baking it to order—each with the following times.

Step	Paula's pizza line	George's pizza line	George's wing line
Prep / dough	4 min	3 min	3 min
Top / season	3 min	2 min	1 min
Cook / fry	4 min	4 min	8 min

The deep fryer also opened up a new possibility: chicken wings. George had grown up frying wings in his family's restaurant, but Paula had no interest in running the fryer herself. If the shop sold wings, George would be the one making them. A finished pizza sold for $12. A finished order of wings sold for $13. They projected that demand for both, at these prices, comfortably exceeded anything the shop could produce.

Paula and George faced a decision about how to use the new kitchen. One option was to keep both stations running pizza: Paula at one, George at the other, each working independently on their own orders. The other option was to dedicate George's station to wings instead, while Paula's station continued to handle pizza alone. For health safety reasons, they felt having a "mixed" line for pizza and wings was a bad idea.

Question for consideration:

> *Should Paula and George run two pizza stations, or one pizza station and one wing station? What should guide their decision?*

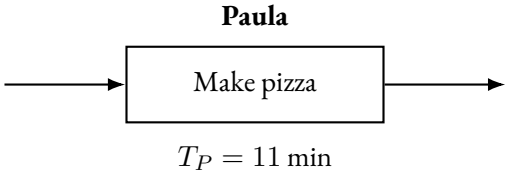
Class discussion

RADU: [*After students finish reading.*] Before we get into the new kitchen, can someone remind us where we left things with Paula and George a few classes back?

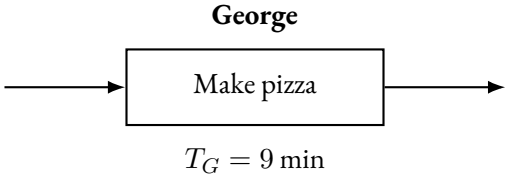
FARAH: They were working in series, one station handing off to the other. The capacity of the whole system was the minimum of their two individual capacities, and whichever of them was slower was the bottleneck.

RADU: Exactly. Now things have changed. Let's first consider the scenario where each runs a pizza station. Can someone draw what Paula's line looks like, on its own?

HAOYU: That part is easy. Paula's looks like this. [*Gets up and draws.*]



And George's looks the same, just with his own flow time.



RADU: And why these numbers?

HAOYU: Well, T_P is the flow time of Paula to make a pizza, which is 4 minutes for prep, plus 3 minutes for top, plus 4 minutes for baking, totalling 11 minutes. A similar calculation gives George's.

RADU: No wait times? Just add up the processing times?

HAOYU: Oh right ... well since they have their own lines, they don't need to wait for each other, but ... oh, maybe they have to wait at the oven?

RADU: You would certainly know better than all of us if that was true.

HAOYU: Actually, let me assume it is an industrial pizza oven. In this case, there are many shelves. I don't think there will be any waiting, the pizzas from the two lines can use the oven independently, I am sure.

RADU: And that means no waiting?

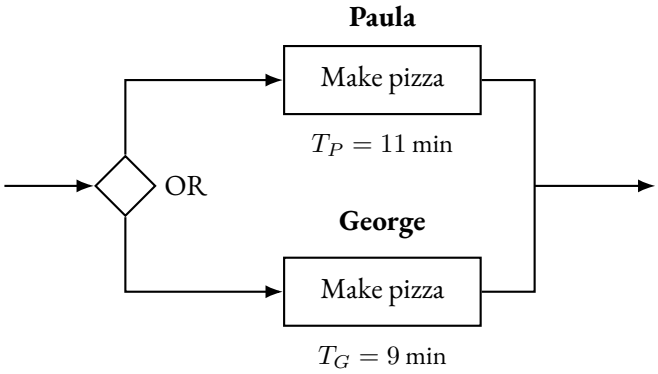
HAOYU: Yes, it's basically like each has its own oven.

RADU: Good. Now here is the question. Orders are not arriving separately to Paula and George. They are arriving to the shop as a single stream of customers, and each customer's order needs to go to one station or the other. How do we put these two pictures together into a single process flow?

[*A pause while everyone looks at the two boxes on the board.*]

FARAH: It's simple, we saw something like this already a couple of times. The order arrives, and then there is a choice of which path it takes. There should be a diamond there. Only one of the two paths gets taken for any given order.

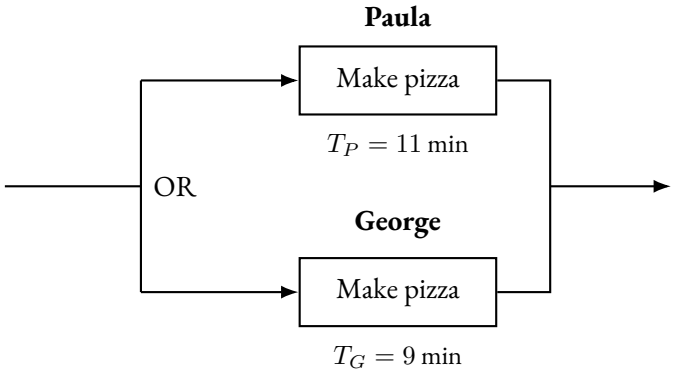
RADU: Let's draw that. [*Turns to the chalkboard.*]



We will call this an *OR split* of the flow. The incoming flow of orders splits into two paths, and any given order travels down *exactly one* of them—*either* to Paula *or* to George—and then the two paths recombine into a single stream of finished pizzas leaving the shop.

MING: I think I am a little confused about the diamond there. I felt there is a diamond when there is some decision being made there on which path the flow moves forward. We saw that with the call center example for where the tickets go to. Here, I feel Paula and George just tackle the next order as it arrives.

RADU: This is a very astute intuition. You know, there are really sophisticated taxonomies and pictures for these process flows. I will send a reference to you via email later. They can be super careful, even pedantic, about these situations. We aren't going to go "all out" here, so when there is no real decision maker or decision rule mapping out the flow, I think it might be better to drop the diamond and just have the flow split, labelling it with this "OR". Is everyone okay with this? [*Redraws the picture.*]

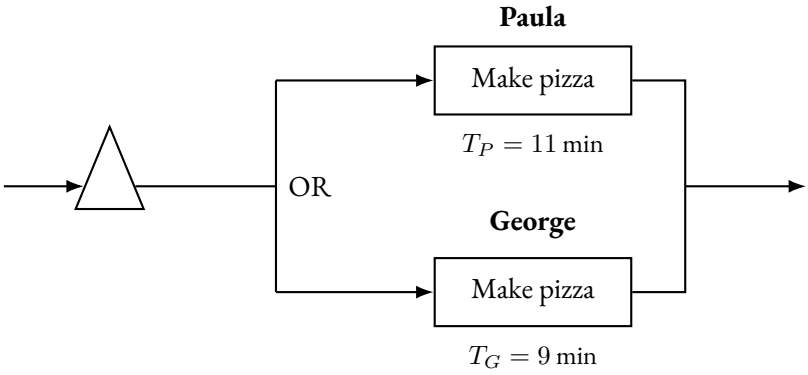


MING: But you are missing something I said. There should be a pile of orders waiting for Paula and George to pick up. Actually, it is something unspecified in the case how these orders are coming in.

RADU: Let's assume there is someone working there to take orders and there is a system to combine these orders with take-out and delivery orders coming by phone or online or whatever. So this pile, is it before or after the split of the flow?

MING: Before. I don't think things should pile up in front of Paula and George in the picture. Orders are for pizzas, not "pizzas made by Paula" or "pizzas made by George".

RADU: Also sensible, let's get the picture right.



Let's take this as our operation. What is its capacity for making pizzas?

MING: [*Answering quickly.*] I want to guess before Farah tells us. [*Everyone laughs, including Farah.*] My first instinct is that it should be some kind of average of Paula's capacity and George's capacity, since orders are splitting between them.

RADU: That guess certainly carries some logic with it. Farah, does it make sense to you?

FARAH: Not fully. Let's just compute their individual capacities first and see. Paula takes 11 minutes per pizza, so her capacity is $60/11$, about 5.45 pizzas per hour. George takes 9 minutes per pizza, so his capacity is $60/9$, about 6.67 pizzas per hour.

RADU: And then, Ming, if we average those?

MING: [*Doing the arithmetic on her phone.*] That would be around 6.06 pizzas per hour.

HAOYU: But at Suzu's we added the two chefs. Two times 30 bowls per hour was 60. We did not average them.

MING: The chefs were the same speed. Paula and George are not. That is why I thought of an average.

FARAH: But that cannot be right. Think about what is actually happening at each station. Paula is making pizzas the entire time, completely independently of George. George is also making pizzas the entire time, completely independently of Paula. Neither of them is waiting on the other for anything. So whatever Paula produces in an hour, and whatever George produces in an hour, both actually happen. The shop's total output potential is not some blend between them. It is both, added together.

RADU: So you are saying the capacity of the system is the sum, not the average, of the two individual capacities?

FARAH: Yes. 5.45 plus 6.67, which is about 12.12 pizzas per hour.

MING: Right ....

CAMILA: That makes sense to me if I think of it as two separate pipes running side by side. Two pipes running in parallel each carry their own flow, and the total water coming out the far end is the sum of both. It is only when water has to pass through one pipe and then another, in series, that the narrower pipe holds everything back.

RADU: I like that image very much. We would say that these pipes are running in *parallel*. This is what you might hear someone call this operation: a *parallel OR operation*. Let's summarize the calculation of capacity across a parallel OR split. [*Writes on the board.*]

$$\boxed{C_{\text{OR split}} = C_{\text{Paula}} + C_{\text{George}} = \frac{60}{11} + \frac{60}{9} \approx 12.12 \text{ pizzas per hour.}}$$

HAOYU: Wait, but if George is faster than Paula, why don't we just have George run both stations? He could go back and forth between them.

RADU: Can he?

HAOYU: [*Pausing.*] Right, of course he can't.

RADU: You really are impressed with George, I can tell that. You almost had him in two places at once! [*Everyone laughs.*]

HAOYU: Yes, he is indeed impressive. But, then again, he is just one person.

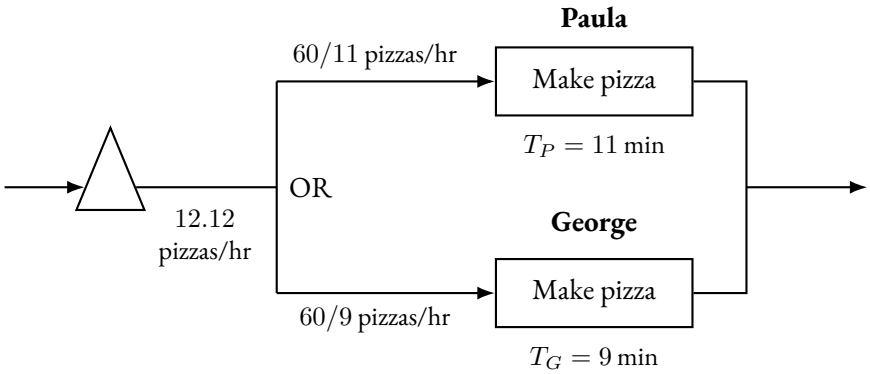
CAMILA: So the extra pizzas do not come from George being fast. They come from there being two separate people, each fully and simultaneously occupied, neither one waiting on the other.

HAOYU: Of course, of course. I don't know what I was thinking.

RADU: I am sure you know this scenario in your bones better than any of us, Haoyu. Formulas and numbers can cloud an issue at first, even an issue you have lived. We are just seeing how all of your experience maps onto these pictures and formulas. Once that mapping is clear, we will see how the formulas may help us. But the fact that things are making sense to you is comforting to me.

HAOYU: Happy to be of service.

RADU: Now, we can put a flow number on each path. It splits like this.



CAMILA: Looks like a good deal for Paula, she is doing less work here?

RADU: What makes you say so?

CAMILA: Well look, Paula handles less flow than George, $60/11$ is less than $60/9$.

RADU: Does that mean she is working less?

CAMILA: She is certainly *doing* less.

RADU: Really? Let me ask you this. Which resource is the bottleneck here?

FARAH: Both are bottlenecks.

MING: Wait a second, both? At Suzu's, Binesi said two resources can both be bottlenecks if they have the same capacity. Paula and George do not have the same capacity.

FARAH: At Suzu's, the kitchen and the stools were on the same path. Recall, this is like two "bottles" after all, as Camila said, two pipes. Each pipe, or each path through the flow, has its own bottleneck.

MING: Oh. [*Pauses.*] Radu, is this right?

RADU: When have you known Farah to be wrong in these situations?

[*Everyone laughs.*]

MING: [*Slowly at first.*] So the idea is that every path of flow has its own bottleneck? [*Pauses.*] Right, yes. I guess that does make perfect sense.

RADU: Every path arising from an "OR" split, yes.

HAOYU: [*Looking upwards, puzzled.*] So wait, you are saying George is a *bottleneck*! He is so much faster than Paula.

RADU: As much as it saddens you to call George a bottleneck, he is one.

HAOYU: But he's so fast!

BINESI: Haoyu, the point is not his speed in comparison to Paula. He is a bottleneck by default: he is the only resource on his line, and so, by definition, he is the bottleneck of that line.

HAOYU: Oh, right. That makes me less upset. [*Everyone laughs.*]

RADU: Every system has at least one bottleneck, since every system has a slowest resource. And today we have learned that every path through the system has its own bottleneck as well. So a system can have *many* bottlenecks, and that is not necessarily a bad thing either.

MING: [*With a smile*]. Well, right now Radu is the bottleneck in moving this discussion forward. And that's not necessarily a good thing.

[*Everyone laughs, including Radu.*]

RADU: Noted. As much as I would like to move on, we got into this little detour based on a comment by Camila. Camila, you said Paula got the better deal here. So let me ask the class directly: who is working harder, Paula or George?

CAMILA: George is. Look at the flows we just drew. Every hour, $60/9$ pizzas leave his station and only $60/11$ leave hers, so more of the shop passes through his hands. If the two of them earn the same wage, George's hour buys the shop more than Paula's hour does.

BINESI: But that just says George is producing more. It does not say he is working harder.

CAMILA: Is there a difference?

BINESI: Of course there is. Paula never stops moving. She tosses, she tops, she bakes, and the moment one pizza leaves her hands she starts the next. George does exactly the same. Her minutes simply turn into fewer pizzas than his do.

FARAH: Each of them is the bottleneck of their own path, so neither of them ever waits on anything. Both are utilized 100% of the time. Measured by how busy they are, they are exactly equal.

CAMILA: Then we have two measures that disagree, and I know which one the bank believes. Capacity is what the shop can actually sell. Whoever contributes more to capacity is doing more work. Effort, on its own, does not pay the rent.

HAOYU: On a Saturday night, nobody in a kitchen asks who has the better deal. You want every pair of hands moving, and both of these pairs are moving.

BINESI: And it is not a competition anyway. As Haoyu points out, two pairs of hands are always better than one. They are a team.

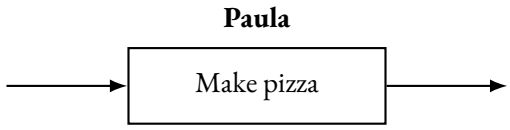
RADU: I think you both make good points. Can we follow Camila's line for a moment and turn to the question of revenue-generating potential. How do we assess that here?

CAMILA: That we can do in one line. Revenue rate equals price times throughput. So the revenue rate for two pizza stations is about 12.12 pizzas per hour times \$12 per pizza.

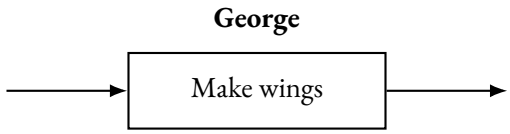
$$\text{Revenue} = \left(\frac{60}{11} + \frac{60}{9}\right) \times \$12 \approx \$145.45 \text{ per hour.}$$

RADU: Good, I like it. Now, let's consider the other setup, where George is dedicated to making wings. How does this system look? [*Offers the chalk he is holding to the class.*]

HAOYU: That one really is simple. There is no split at all now. Pizza orders only ever go to Paula, and wing orders only ever go to George. So again, we have these two lines, completely disconnected from each other. [*Gets up, takes the offered chalk from Radu, and draws.*]



$T_P = 11$ min, $C_P = 5.45$/hr



$T_G = 12$ min, $C_G = 5.00$/hr

RADU: I like that you wrote the capacity of each underneath. How did you get $C_G = 5$?

HAOYU: There are 3 minutes prep, 1 minute seasoning, 8 minutes frying. 12 minutes total, so a capacity of $60/12$, exactly 5 orders per hour.

RADU: And so what is the revenue in this system?

FARAH: Pizza revenue is $\frac{60}{11} \times \$12 \approx \$65.45$ per hour. Wing revenue is $5 \times \$13 = \$65.00$ per hour. Together,

$$\text{Revenue} = \$65.45 + \$65.00 = \$130.45 \text{ per hour.}$$

This assumes, of course, that both stay maximally busy, which is the exercise we are undertaking here since we are looking at revenue-generating potential.

RADU: So, comparing the two options?

FARAH: Two pizza lines wins: $\$145.45$ against $\$130.45$. About \$15 an hour more, which is over 11%.

RADU: Making sense to everyone?

[*A long pause.*]

CAMILA: I don't have a question here, but more of a comment. I think an important reason we can do this type of calculation is that we assume the demand is sufficient in both cases. So whatever Paula and George make, they can sell all out. But in practice, I am not sure if that really holds. In that case, our calculation like this would have failed.

RADU: Very sharp observation, Camila. Yes, our calculation do require certain assumptions. When we talk about capacity, we are thinking about how much demand we can handle, assuming we can get as much demand to come as we want. That is what it means to talk about capacity. But your concern is reasonable Camila. Suppose instead of assuming sufficient demand, I told you the demand for wings was less than what George makes, is there a way to adjust our calculation here?

[*Another long pause.*]

FARAH: Suppose instead of 5 wings per hour, the real demand for wings is 3 per hour. George can just fry 3 wing set instead of 5 per hour, and the revenue from wings would be $\$13 \times 3 = \$39$.

CAMILA: Ah, I see. It's like what actually sells is the minimum of George's capacity and real demand. Then it does fix our calculation. Thanks, Farah.

MING: Can I go back to something Haoyu said? [*Turning to Haoyu.*] You called these two lines completely disconnected. Is that actually true?

HAOYU: What do you mean? Paula makes pizza, George makes wings. Nobody hands anybody anything.

MING: Sure, but that is only true as long as every customer orders one thing *or* the other. What if someone at the counter wants a pizza *and* a side of wings, as a single combo order? Then suddenly Paula and George are not so disconnected, are they? The customer is waiting on both of them.

RADU: Ah ...right. That is a sharp thing to notice, Ming, I overlooked it. How should this change the process flow?

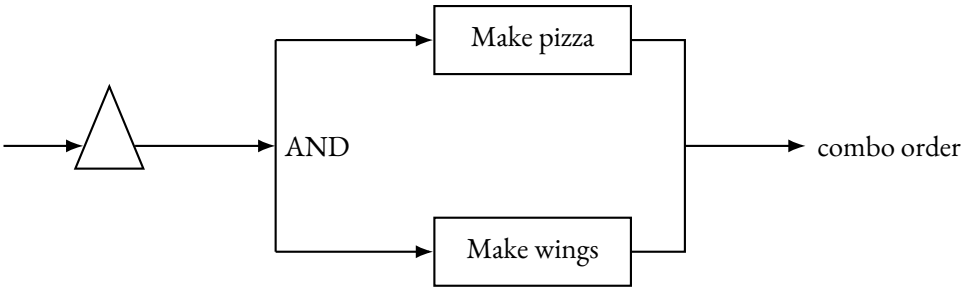
[*A pause while the class thinks it through.*]

CAMILA: That does not sound like an OR split anymore. The combo order would not go to Paula or to George. It would need to go through both.

FARAH: That is an "and" instead of an "or".

RADU: Can you draw it?

FARAH: Sure. [*Takes the chalk.*]

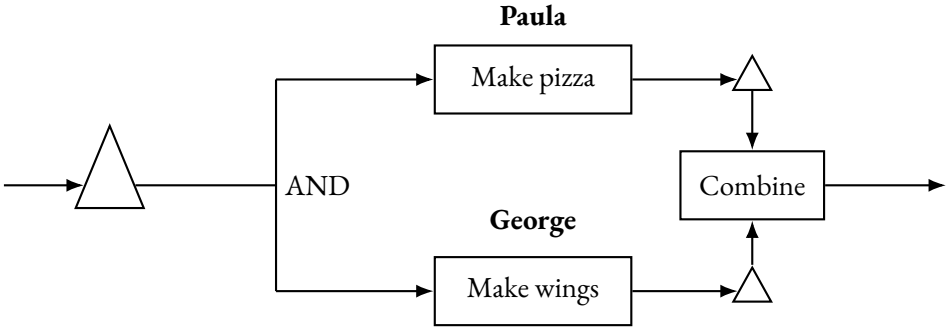


An order is not complete, and cannot leave the shop, until both the pizza and the wings are ready. Paula and George can work on their part simultaneously. This part is the same as before. But the order has to wait for whichever of them finishes last and then it is assembled into a combo.

BINESI: So doesn't that mean there is some waiting later in the picture now?

RADU: How so? [*Offers Binesi some chalk.*]

BINESI: [*Comes to the board.*] Well, Farah just said that completed pizzas or wings might need to wait to be paired up in a combination. So I think it should look something like this. [*Adjusts the figure.*]



RADU: That, indeed, looks more accurate. But I can say that figures of the kind Farah drew are common to see. When we have these "and" splits, little triangles like this are somewhat implied.

BINESI: That makes sense.

RADU: We call these splits *AND splits* or *parallel AND splits*. The resources are working in parallel, but they are not totally independent, in the sense that their outputs are combined. So what is the capacity of this AND-split system?

MING: [*Confidently, but then catching herself.*] I wanted to say we add the two capacities again, since they are working at the same time, but that does not sound right. They are not producing separate finished things anymore. They are producing two halves of the same thing.

RADU: And so?

MING: [*Pauses and then answers carefully.*] It feels more like the old serial system. The combo order cannot leave until the slower of the two is done.

FARAH: Right! It behaves like a serial system with respect to capacity, even though Paula and George are working at the same time rather than one after the other. The combo capacity is the minimum of the two, not the sum.

RADU: Like this? [*Radu writes on the board.*]

$$\boxed{C_{\text{AND split}} = \min\{C_{\text{Paula}}, C_{\text{George, wings}}\} = \min\left\{\frac{60}{11}, \frac{60}{12}\right\} = \frac{60}{12} = 5 \text{ combos per hour.}}$$

FARAH: Yes, precisely.

RADU: And so what happens to the revenue now?

CAMILA: The case never tells us what a combo would sell for, so let me just assume it is priced as the sum of its parts, no bundle discount and no markup. $12 for the pizza plus $13 for the wings is $25 per combo. At a capacity of 5 combos per hour, that gives

$$\text{Revenue}_{\text{AND}} = 5 \times \$25 = \$125 \text{ per hour.}$$

RADU: And how does that compare to the other two configurations we found today?

CAMILA: It is actually the worst of the three. Less than the two pizza lines at $\$145.45$, and less than pizza and wings sold separately at $\$130.45$.

RADU: Does that surprise you?

CAMILA: Not really. Paula is not fully used in the "and" case. She is fully utilized in the other two. Her capacity is $60/11$, about 5.45 pizzas per hour, but the combo can only move at George's pace of 5 combos per hour. She has nearly half a pizza's worth of capacity per hour that simply goes unused, waiting on George.

RADU: So are both bottlenecks still, like in the parallel-OR system?

FARAH: No, George is the only bottleneck now.

HAOYU: [*As if woken up suddenly from a daydream.*] Wait, George?! The *only* bottleneck!

FARAH: Yes, George is the slower of the two resources in terms of capacity, and both need to process each combo.

HAOYU: Oh, so we are saying that all customers order both pizza and wings as a combination in this scenario?

RADU: Yes, that is what this last case was.

HAOYU: Right, then it makes sense. In reality, people can order pizza only or wings only, or multiples of each, right?

RADU: Of course they can, but let's not make our lives so complicated right now.

MING: [*As an aside but loud enough for everyone to hear.*] Radu, as usual, copping out, not treating the real problem.

RADU: [*A little frustrated.*] What is real here? This is a fictional story! Can I remind you of the premise? We are just trying to learn some concepts here we can use in real life. If we had the full complexity of real life in class, then we would all be running a pizza restaurant right now. Do you want to do that?

MING: Hell no! [*Pauses for a second.*] No offense Haoyu.

HAOYU: None taken. [*Turns from Ming to Radu.*] But there is one real-life complexity that does bother me, sorry. Why is George the one making wings at all? He is the better pizza maker: 9 minutes compared to Paula's 11. If one of them has to leave pizza behind, should it not be Paula?

BINESI: But Paula doesn't want to do it. The case says she has no interest in running the fryer.

MING: And even if she did, Haoyu, your argument is missing a number. Knowing that George is better at pizza than Paula tells us nothing about who should make the wings. For that, we would need to know how good Paula is at wings, and the case never says.

RADU: Suppose the case did tell us. What would we compare?

CAMILA: How much pizza each of them gives up for every order of wings they make. Whoever gives up less pizza makes the wings. Even if Paula were slower than George at both products, pizza could still rightly be hers if her wing making was really, really slow.

RADU: Economists call that *comparative advantage*.

BINESI: Whatever we call it, it doesn't matter. Paula doesn't want to make wings, end of story.

CAMILA: Paula doesn't want to make money?

RADU: [*Raising a hand before Binesi can answer.*] Okay, okay, let's not go down this rabbit hole again with you two. What people want to do and what the numbers say they should do is a real tension, and it deserves more than the five minutes we have left. For now, the case asked us about two configurations. Where do we stand on those?

FARAH: So, with respect to the original question, we can say that the system with two dedicated pizza lines is better. That gives more revenue.

RADU: Is everyone comfortable with that?

[*Binesi raises her hand slowly.*]

BINESI: I am, mostly. But something is bothering me. If they go with two pizza stations, what happens to the deep fryer? It would sit there completely unused, that seems like a waste.

CAMILA: That should not factor into the decision. The fryer is a sunk cost. They got the new place and it was already in there, whether they use it or not. The only question that matters now is which configuration earns more going forward, and the math says two pizza stations.

BINESI: I understand that argument, and I am not disagreeing with the numbers here. But it still does not sit right with me. If they are going to own this thing, sitting in their kitchen, taking up space, shouldn't they actually use it for something?

CAMILA: Why? Owning something does not obligate you to use it. That is exactly the trap of sunk-cost thinking—feeling pressure to use a resource just because you already paid for it, rather than because using it is actually the best decision today.

BINESI: [*Thinking out loud.*] Maybe it is not about whether *they* use it, though. Maybe it is about whether the fryer gets used at all, by anyone. If they are not going to use it, perhaps the answer is not to let it sit there idle. Perhaps the answer is to sell it? There might be another kitchen nearby that actually needs a deep fryer and does not have one yet. That way the equipment still ends up doing something useful. It is only wasteful if it sits there gathering dust forever, not if it goes to wherever it is actually needed.

RADU: So whether Paula and George use the fryer, and whether anyone uses it, are two different questions?

BINESI: Right. I think I was conflating them. I feel better about it now.

CAMILA: And maybe they keep it to use it later, maybe they expand into wings at some point down the line.

RADU: Another good idea. Shall we wrap up there?

MING: [*Interjects.*] Can I ask something before we close this out? What about the flow time in all of this, the "and" versus 'or"? How long does any particular customer actually wait for their pizza? That also seems important.

RADU: A very good question, and one I want to deliberately set aside today. We have only been talking about capacity and revenue. Flow time in a system with splits like this has its own subtleties, and I would rather give it the attention it deserves in a class of its own, rather than rush it now.

MING: Fair enough. I am tired of this class today anyway.

[*Everyone laughs.*]

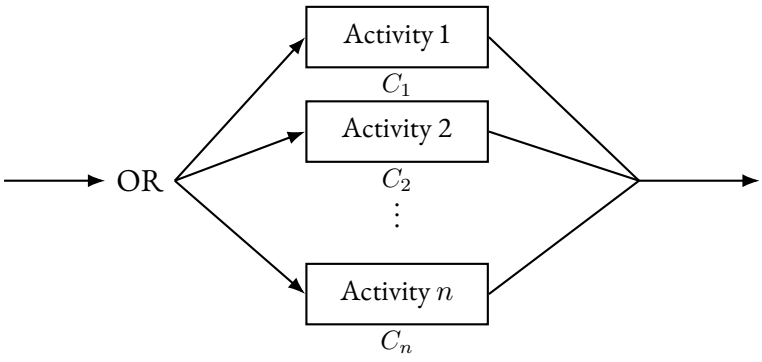
RADU: Ming, we can always rely on you to leave us on a cheerful note.

Email

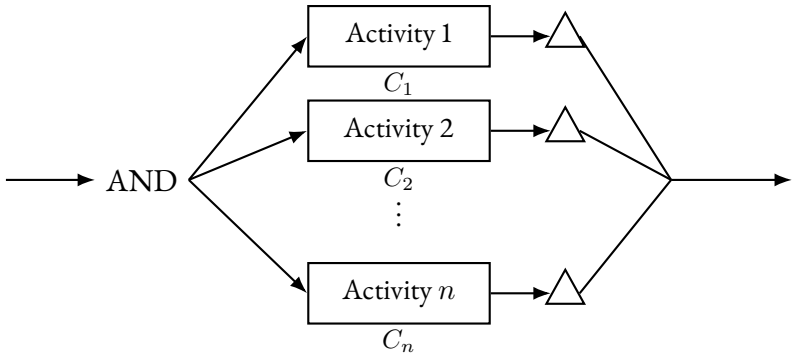
Dear students,

Thank you for today. For several classes now we have lived inside serial systems, where work passes from hand to hand along a single line and the slowest resource sets everyone's pace. Today the line split in two, and the split turned out to have two very different characters. Let me consolidate the formulas from today and then develop a few of the threads that ran underneath them.

OR splits and AND splits. Recall the two structures we worked with today. In an *OR split*, an order travels down *exactly* one of several paths. The two chefs at Suzu's were already an example: a resource pool is an OR split whose paths happen to be identical, which is why we could simply multiply 30 bowls per hour by 2. In a general setting of n activities, an OR split looks like this:



In an *AND split*, an order requires every path to finish before it is complete. In a general setting of n activities, an AND split looks like this (I have drawn in Binesi's little waiting triangles before the paths recombine, since finished parts may wait for their partners):



The capacity formulas could not be more different in spirit, even though the pictures look superficially similar:

$$\boxed{\begin{aligned} \text{OR split:}\quad & C_{\text{system}} = C_1 + C_2 + \cdots + C_n \\ \text{AND split:}\quad & C_{\text{system}} = \min\{C_1, C_2, \ldots, C_n\} \end{aligned}}$$

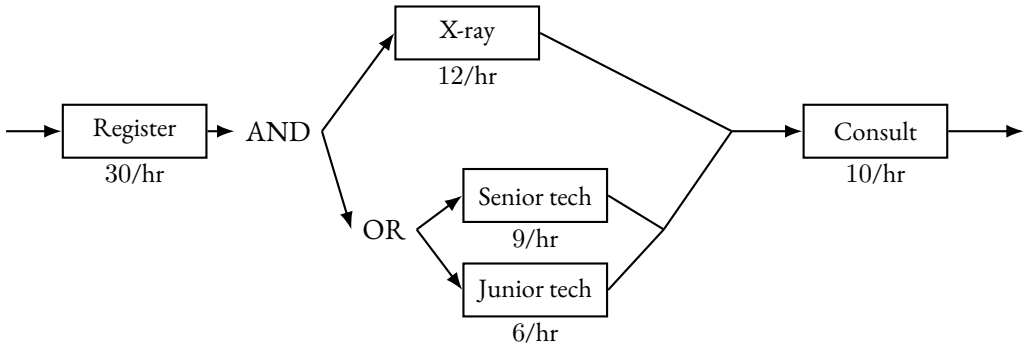
In an OR split the capacities add, and each path keeps a bottleneck of its own; in an AND split the slowest resource binds, just as in a serial system. The reason is worth restating, because it is easy to memorize the formulas without internalizing why they differ. In an OR split, the paths never interfere with each other—whatever Paula produces and whatever George produces both actually happen, as Farah put it in class, so the system's total output is the sum. In an AND split, every unit of output requires *all* paths to have done their part, so no matter how fast one path is, the system can only finish units as quickly as its slowest path allows. This is the same logic as a serial system, just executed simultaneously rather than sequentially.

You may already notice that the AND Split and the serial process feels quite similar in spirit. Then why distinguish these two? A simple answer is that the serial process and an AND split differ in whether activities can happen simultaneously in the same order. For example, with separate workers taking 11 minutes to make a pizza and 12 minutes to make wings, preparing a combo *sequentially* takes 23 minutes, whereas preparing both items simultaneously takes 12 minutes, ignoring waiting and assembly time. Another name for processing simultaneously is to say the resources work *in parallel* on the *same* order. Yet both arrangements have the same capacity of 5 combos per hour, limited by the wings station. Why? Even in a serial process, workers can work simultaneously on *different* orders: while George makes wings for one order, Paula can make the pizza for the next.

Complicating the story of bottlenecks. In my email after our first class with Paula and George, I mentioned that in operations with multiple paths there can be a different bottleneck on each path, and promised we would return to it. Today we did, and I want to make sure to reiterate the lesson, because it changes how we should speak. Back in the serial world, it was largely harmless to talk about *the* bottleneck of a system, the way we talk about the weakest link of a chain. That habit of speech should now feel wrong to you (even though you, like me, will still do it without thinking). Every path created by an OR split carries its own bottleneck—its own slowest resources. A system with many paths has many bottlenecks, and this is not a defect of the system; it is simply what the word means once flow is allowed to branch.

Let me make that concrete with an example away from the kitchen, one that Ming might recognize. Picture a small walk-in clinic. Every patient first registers with a receptionist, who can handle 30 patients

per hour. Every patient then needs two things before seeing the physician: an X-ray, and blood work. The imaging room can process 12 patients per hour. The blood draw can be done by either of two technicians: the senior one manages 9 patients per hour, the junior one handles 6 per hour. In other words, the blood station is an OR split sitting inside one branch of an AND split. Finally, every patient sees the physician after all tests are complete, who takes 6 minutes per consult, for a capacity of 10 per hour.



We evaluate the system from the inside out. The innermost structure is the OR split at the blood station: capacities add, giving $9 + 6 = 15$ patients per hour. Similar to the pizza store example, both technicians are bottlenecks in this OR split.

Now we go one level out and look at the AND split with the imaging room: capacities combine by the minimum, giving $\min\{12, 15\} = 12$. So now the imaging room becomes the bottleneck.

Finally, let's look the the whole picture. That AND split sits inside a serial chain with the receptionist and the physician: the minimum again. Altogether,

$$C_{\text{clinic}} = \min\Big\{\, 30,\ \min\{12,\ 9+6\},\ 10 \,\Big\} = 10 \text{ patients per hour.}$$

Now the physician is the bottleneck of the current system!

I hope this example updates your understanding about the concept of a bottleneck. A bottleneck is not a title held by one resource in the building. When someone tells you their operation's bottleneck, it is fair to ask: the bottleneck of which path, under which setup, as of when?

Scale versus scope. The actual decision Paula and George faced today about whether to have two pizza stations or one pizza station and one wing station has a name in the strategy and operations literature. Two pizza stations is a bet on *economies of scale*: more capacity devoted to the same product, hoping that doing more of one thing is more profitable than diversifying. One pizza station and one wing station is a bet on *economies of scope*: using shared infrastructure (the same kitchen, the same brand, the same customer base) to support more than one product line, hoping that variety creates value the single-product approach cannot.

It is worth noticing that today's numbers favored scale over scope, but this is not a general law. We made the assumption that there was always sufficient demand for both pizzas and wings to consume all of the restaurant's capacity, but it might be that demand for pizza alone would lag, and that offering wings would bring in customers that pizza alone does not. The right balance between scale and scope thus depends on demand patterns we did not model today. A real owner would want to think about these demand-side effects too before generalizing today's conclusion.

Why bundling can destroy value. A striking number from today was not the comparison between the two OR configurations (the two pizza lines against the independent pizza and wing lines), but how badly the AND-split "combo" setting performed. Forcing every order through both Paula and George simultaneously gave the lowest revenue of the three options we considered.

This is not a quirk of our particular numbers. Rather, it is a small instance of a real and well-studied phenomenon in economics. William Adams and Janet Yellen, in a 1976 paper that remains a standard reference on the subject, show formally that bundling two goods together can sometimes *reduce* a seller's revenue relative to selling them separately, because bundling forces every customer to take both goods at a single combined price, even customers who would have happily paid for one and not the other [1].

From the operational perspective, bundling destroys value by reducing the system ' s flexibility. When products are sold only as a bundle, the process has an AND requirement where all components must be ready before the order can move on. By contrast, when customers can buy either item separately, orders can follow an OR split where each order can be completed once its chosen branch finishes, without waiting for the other branch. In comparison, bundling adds a constraint by tying the products together in fixed proportions. This creates waiting and prevent useful capacity from earning revenue. Thus the operational cost comes from this loss of freedom.

A note on diamonds, OR, AND, and OMG. The difference of opinion between Farah and Ming in class, about when a split deserves a diamond, deserves a slightly more careful answer than I gave on the spot. The short version, which I will not belabor here because it starts to split hairs we mostly want to avoid in this course: a real industry standard for this kind of diagramming exists, called BPMN (Business Process Model and Notation), maintained by the Object Management Group (OMG). BPMN distinguishes our two split types more finely than we have—an *exclusive* gateway (our diamond, a genuine condition that picks one path), a *parallel* gateway (our AND split), an *inclusive* gateway (one or more paths, each independently conditional), and an *event-based* gateway (the path taken depends on which of several uncertain events happens first, rather than on any condition you could check in advance). If you want to go further with this notation than we will in this course, check out the formal specification maintained by the OMG. For our purposes, the simpler distinction we have been using including a diamond when we want to emphasize the role of a condition impacting the choice of flow, a branch labeled "OR" or "AND", will continue to serve us well, and I do not think we need the full BPMN taxonomy to think clearly about the processes we will study.

Warmest, Radu.

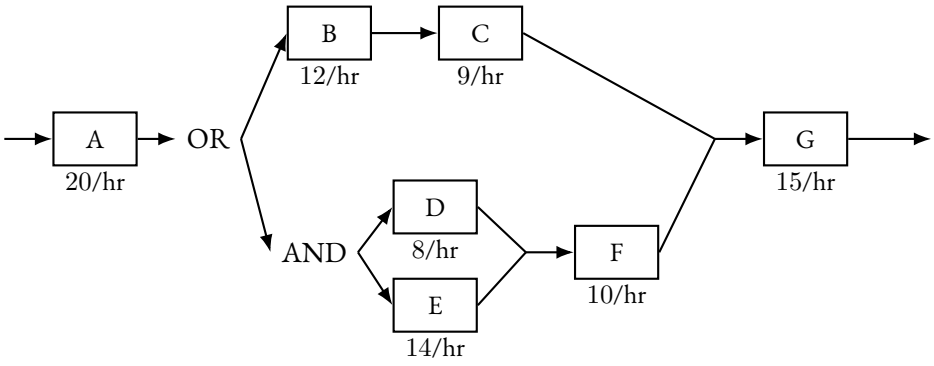
References

[1] William James Adams and Janet L. Yellen. Commodity bundling and the burden of monopoly. *Quarterly Journal of Economics*, 90(3):475–498, 1976.

Extensions

[*Posted as homework on the course's website for this class session.*]

1. Below is a process flow slightly more tangled than Paula and George's. Activities are labeled with their resource's capacity in units per hour.



   (a) Compute the capacity of each path and of the system as a whole, and identify every bottleneck in the picture—the system's, and each path's.

   (b) Write the system's capacity as a single nested formula in $C_A, \ldots, C_G$, the way the email evaluates the clinic. Convince yourself the formula can be read directly off the diagram.

   (c) You can afford to speed up exactly one resource. Which one do you choose, and how much additional system capacity can that single investment buy before the binding constraint moves somewhere else?

   (d) Think of an example of what the activities A, B, C, D, E, F, and G could be that would "bring to life" this process flow.

2. Draw your own process flow diagram with at least eight activities that uses at least one OR split, at least one AND split, and at least one serial chain—arranged so that the system's bottleneck is the resource with the *largest* individual capacity anywhere in your diagram. If you believe this is impossible, prove it; if you manage it, say what your construction teaches about judging a resource by its individual capacity.

3. At Paula and George's pizza place, suppose the person at the till (not mentioned in the case) routes orders by strict alternation: first order to Paula, second to George, third to Paula, and so on, no matter who is free. Explain why the shop's capacity drops under this scenario and explain where the lost pizzas go. Use this conclusion to argue why Ming's insistence that the pile sits before the split matters.

4. Camila assumed a combo sells for $25, the sum of its parts.

   (a) What combo price would the AND-split configuration need to charge to match the $145.45 per hour of the two pizza lines?

   (b) In practice, combos are usually priced *below* the sum of their parts, not above. Combining your answer in (a) with the email's discussion of bundling, explain to Paula and George why the all-combo case is doubly bad.

(c) Find a real combo on a real menu this week. Is it priced above or below the sum of its parts? Offer a guess at what the restaurant knows about its own kitchen, or its own customers, that makes that price sensible.

5. Ming pointed out that Haoyu's argument was missing a number: how good Paula is at wings. Suppose we learn it: Paula would take 4 minutes of prep, 2 minutes of seasoning, and the same 8 minutes of frying per order of wings.

   (a) There are now four ways to staff the two dedicated stations: both on pizza, both on wings, George on wings with Paula on pizza, and Paula on wings with George on pizza. Compute the hourly revenue of all four and rank them.

   (b) What does this tell you about the comparative advantages of Paula and George?

6. Camila opened one debate today by claiming Paula "is doing less," and the class never settled it. Compute each person's utilization in all three configurations from class: the two pizza lines, the dedicated pizza and wing lines, and the all-combo AND split. In which configuration does someone finally get idle time, who is it, and what does your answer do to Camila's claim?

7. A used-equipment dealer offers Paula and George $1,800 for the deep fryer. Camila called the fryer a sunk cost. Is it still one? Say precisely what, if anything, the offer changes about the decision to run two pizza lines versus one pizza line and one wing line.

8. Think of the last group assignment you completed, in this program or on any team. Draw the process flow of how the work actually moved from the assignment being handed out to its submission. Mark each split as OR or AND, and each handoff in series. Where was the bottleneck of each path? Is the flow you drew the one your group planned?

Reflections

[*Posted as a reflection exercise on the course's website for students to write about in a journal due at the end of semester.*]

1. Ming joked that Radu was the bottleneck holding up the discussion, and everyone laughed, including Radu. Could you make that joke to the person setting the pace where you work? What does your answer tell you about the place?

2. Binesi conceded every number and still could not accept the fryer sitting idle. What do you own that you keep only because letting it go would feel like admitting the money was wasted?

3. Camila asked "Paula doesn't want money?" as if that settled the matter. Recall a time, if any, when you turned down work that could have paid well. How would you explain that choice to someone who thinks like Camila?

4. In the work you do with others, are you more often an OR path, where someone else could have done your part instead, or an AND path, where nothing is finished until your part is? (If neither fits your current role, think of one that did.) How does each one feel from the inside?

5. Are you somebody else's "sunk cost": someone they keep investing in partly because they already have? Is that a good feeling or a bad feeling? How would you tell whether they keep investing for that reason, or because they expect something back?